Highlights

Delay-Integrated Cascading Failure Analysis in Cyber-Physical Power Systems: A Closed-Loop Feedback Perspective

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- A closed-loop cascade framework embeds communication delay as an endogenous causal variable inside each cascade round.
- Delay damage is asymmetrically concentrated at moderate-to-high tolerance margins where delay-free models predict safety.
- Delay amplifies stochastic variability, creating heavy-tailed collapse distributions absent under ideal communication.
- Generator dispatch deviation metric R_3 reveals hidden functional degradation invisible to aggregate load-retention metrics.
- Round-level heatmap analysis identifies Round 2 as the critical intervention window for delay mitigation.

Delay-Integrated Cascading Failure Analysis in Cyber-Physical Power Systems: A Closed-Loop Feedback Perspective

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ARTICLE INFO

Keywords:
Cyber-physical power system
Cascading failure
Communication delay
Closed-loop feedback
Robustness assessment
Interdependent networks

ABSTRACT

Cascading failure models for cyber-physical power systems (CPPS) routinely treat communication delay as a post-hoc observation rather than a causal driver of further damage. This architectural choice eliminates the mechanism through which delay inflicts its greatest harm: a closed feedback loop in which degraded communication reduces generator controllability, the resulting power imbalance triggers additional branch overloads, the consequent structural failures further damage the communication topology, and the damaged topology amplifies delay for the next cascade round.

In this paper we embed communication delay directly into the cascading failure process. At each cascade round, a delay-dependent control efficiency factor is computed for every surviving generator and applied before the DC power flow that determines overloads. End-to-end latency is decomposed into uplink measurement delay and downlink execution delay through a role-aware model that distinguishes control centers from ordinary relay nodes. Interdependent coupling between the IEEE 39-bus power grid and a Barabási–Albert scale-free communication network is established via a betweenness-assortative strategy, with control centers placed using a composite structural importance score.

Monte Carlo experiments spanning 1 000 independent coupling realizations, eleven tolerance levels, and five delay-severity scenarios yield three principal findings: (i) the delay penalty concentrates asymmetrically at moderate-to-high tolerance margins—precisely where delay-free analysis predicts safety but delay-burdened systems collapse; (ii) delay amplifies the stochastic variability of cascade outcomes, transforming narrow distributions under ideal communication into heavy-tailed distributions under severe delay; and (iii) generator dispatch deviation remains stratified across delay scenarios even when aggregate load retention appears adequate, exposing hidden functional degradation invisible to load-based metrics alone. These results demonstrate that delay-free robustness models are not merely imprecise but systematically misleading, providing the greatest reassurance in operating regimes that harbor the greatest vulnerability.

1. Introduction

The deep integration of physical infrastructure with communication networks has transformed traditional power grids into cyber-physical power systems (CPPS), in which a physical power layer and a cyber communication layer operate as a tightly coupled whole [14, 12]. Under normal conditions, this coupling improves dispatch flexibility and situational awareness. Under stress, however, it creates a bidirectional failure channel: a disturbance originating in either layer can propagate to the other, triggering cascading failures that neither layer would experience in isolation [4, 7].

A rich body of literature has investigated cascading failures in coupled networks through topological models, flow-redistribution rules, and attack-vulnerability analyses [11, 6, 3]. These studies have established how structural interdependence amplifies failure propagation. Yet most of them treat the communication layer as a binary entity—a link is either functional or failed—without modeling the *quality* of communication service that surviving links provide. In particular, communication delay, the time elapsed between issuing a measurement or command and its arrival at the destination, is largely absent from cascading-failure analysis. When delay is mentioned, it is typically reported as a post-hoc observation ("delay increased after failures") rather than modeled as a causal variable that directly degrades generator control performance during the cascade itself [10, 8].

This omission matters more than it might first appear. In a closed-loop control architecture, stale measurements cause the control center to make decisions based on outdated system states, while lagged command execution means that corrective actions arrive after the system has drifted further. Both effects reduce the effective generation output relative to its intended reference. When this reduction is embedded directly into the cascading failure process—before the power-flow redistribution that determines which branches overload—a self-reinforcing feedback loop emerges:

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> *Delay reduces generation → altered power flows cause additional overloads → overloaded branches trip → structural failures degrade the cyber topology → degraded topology increases delay → further output reduction → ⋯*

This feedback loop transforms communication latency from a passive symptom of system distress into an active driver of further damage. Crucially, it is invisible to models that apply delay as a post-hoc correction, in which all delay scenarios share an identical structural cascade and delay merely rescales final performance metrics [9].

Our prior work [5] established a CPPS cascading failure framework with distributed control centers, a Functional Viability-Constrained Island Detection (FVC-ID) mechanism, and a Cohesion-Propagation Importance Metric (CPIM) for optimal control-center placement. That study did not, however, model communication delay's influence on generator control during the cascade. The present paper fills this gap.

The contributions of this work are as follows:

1. **A delay-integrated cascading failure framework with closed-loop feedback.** We embed delay computation directly into each cascade round, scaling generator output before DC power flow redistribution. Each delay scenario therefore produces a genuinely different cascade trajectory—different overload patterns, different failure sequences, different terminal states—rather than a rescaled version of a common trajectory.
2. **A bidirectional, role-aware delay model.** Uplink (measurement) and downlink (command) delays are separately computed from link-level serial and propagation components, with role-dependent service delays for control centers versus ordinary relay nodes. This captures the asymmetric processing behavior inherent in SCADA-type control architectures.
3. **Mechanism decomposition into two degradation channels.** We decompose delay-induced robustness loss into continuous efficiency erosion (path lengthening reduces η) and discontinuous reachability collapse (cyber-network fragmentation severs all CC access, setting $\eta = 0$).
4. **Identification of the asymmetric delay-penalty regime.** Through a delay-penalty heatmap over the (tolerance, cascade-round) plane, we demonstrate that delay damage is largest at moderate-to-high tolerance margins—where the delay-free system retains most load but the delay-burdened system collapses—revealing that delay-free models create a false sense of security precisely where operators believe the system is adequately protected.

The remainder of this paper is organized as follows. Section 2 presents the CPPS model, including the communication delay formulation and interdependent coupling. Section 3 describes the delay-integrated cascading failure dynamics. Section 4 reports experimental results. Section 5 concludes the paper.

2. Cyber-Physical Power System Model

The system under study consists of two interdependent network layers—a physical power grid and a cyber communication network—whose coupling transforms isolated disturbances into cross-domain cascading failures.

2.1. Power Grid Layer

The physical layer is represented by the IEEE 39-bus (New England) test system, a well-established benchmark comprising 39 buses, 46 transmission branches, and 10 generators. Bus connectivity is encoded in an adjacency matrix A_P extracted from the branch table, and the initial operating state—bus loads, generator outputs, and branch power flows—is obtained by solving a DC power flow on the full network [16].

Under the DC power flow assumption, the active power injections satisfy

$$\mathbf{P} = \mathbf{B}\boldsymbol{\Theta}, \tag{1}$$

where $\mathbf{P} = [p_1, p_2, \ldots, p_n]^T$ is the nodal injection vector ($p_i > 0$ for generators, $p_i < 0$ for loads), $\boldsymbol{\Theta}$ is the bus voltage-angle vector, and $\mathbf{B}$ is the bus susceptance matrix with elements

$$B_{ij} = \begin{cases} -b_{ij}, & i \neq j, \\ \sum_{k \neq i} b_{ik}, & i = j, \end{cases} \tag{2}$$

where $b_{ij} = 1/x_{ij}$ is the reciprocal of the reactance of the line connecting buses i and j. Each branch e carries an initial power flow $P_e^{(0)}$ that serves as the reference for the capacity model in Section 2.4.

2.2. Communication Network Layer

2.2.1. Scale-Free Topology

The cyber layer is constructed as a Barabási–Albert (BA) scale-free network to capture the hub-dominated, heterogeneous degree distribution commonly observed in communication infrastructures [2]. The network contains $|V_C| = |V_P| + n_{\text{cc}}$ nodes, where $n_{\text{cc}} = \lceil 0.2 \times |V_P| \rceil$ additional nodes serve as designated control centers (CCs). For the IEEE 39-bus system this yields $|V_C| = 39 + 8 = 47$ nodes, of which 8 are CCs.

The BA growth process starts from a fully connected seed of $m_0 = 4$ nodes. At each step a new node is added and connected to $m_e = 2$ existing nodes selected with probability proportional to their current degree (preferential attachment), producing a power-law degree distribution characteristic of real communication networks.

2.2.2. Control Center Placement via MSIS

Rather than selecting CCs by a single centrality measure, we employ a Multi-attribute Structural Importance Score (MSIS) that fuses complementary topological features:

$$w_i = \frac{\widetilde{B}(i) + K_s(i)}{C_l(i)} + D(i) \cdot C(i), \tag{3}$$

where $\widetilde{B}(i) = B(i) + 1$ is the shifted betweenness centrality, $K_s(i)$ is the k-shell index, $C_l(i)$ is closeness centrality, $D(i)$ is degree, and $C(i)$ is the local clustering coefficient. The top-n_{cc} nodes ranked by w_i are designated as control centers.

This composite score balances global bridging importance (betweenness, closeness) with local structural embeddedness (degree, clustering, k-shell). Our prior work proposed the Cohesion-Propagation Importance Metric (CPIM) for the same purpose [5]; MSIS adopts a similar multi-attribute philosophy while using a different functional form suited to the delay-focused analysis of this paper.

2.2.3. Communication Delay Model

Communication delay is modeled at the individual link level and aggregated over multi-hop paths, with explicit directional asymmetry between uplink (measurement) and downlink (command) traffic.

Link-level delay. For a single surviving link (u, v), the per-hop delay is

$$T_{\text{link}}(u, v) = \frac{S}{R} + \frac{d}{V}, \tag{4}$$

where S is the packet size (bits), R is the link data rate (bps), d is the effective link distance (km), and V is the signal propagation speed (km/s). The first term captures serialization delay; the second captures propagation delay. Uplink and downlink traffic use different packet sizes—measurement packets ($S_{\text{up}} = 8\,192$ bits) are larger than command packets ($S_{\text{down}} = 2\,048$ bits)—introducing directional asymmetry even over the same physical link.

Path-level delay. For a multi-hop path $\pi = (u_0, u_1, \ldots, u_k)$, the total path delay aggregates per-hop link delays, endpoint service delays, and intermediate forwarding delays.

Downlink (CC $\to$ non-CC):

$$T_{\text{down}}(\pi) = \sum_{i=0}^{k-1} T_{\text{link}}^{\text{down}}(u_i, u_{i+1}) + \tau_{\text{CC}}^{\text{tx}} + \tau_{\text{nonCC}}^{\text{rx}} + \sum_{j=1}^{k-1} \tau_{\text{fwd}}(u_j). \tag{5}$$

Uplink (non-CC $\to$ CC):

$$T_{\text{up}}(\pi) = \sum_{i=0}^{k-1} T_{\text{link}}^{\text{up}}(u_i, u_{i+1}) + \tau_{\text{nonCC}}^{\text{tx}} + \tau_{\text{CC}}^{\text{rx}} + \sum_{j=1}^{k-1} \tau_{\text{fwd}}(u_j). \tag{6}$$

The forwarding delay at each intermediate node is role-dependent:

$$\tau_{\text{fwd}}(u) = \begin{cases} \tau_{\text{CC}}^{\text{fwd}}, & \text{if } u \text{ is a CC}, \\ \tau_{\text{nonCC}}^{\text{fwd}}, & \text{otherwise}. \end{cases} \tag{7}$$

Control centers are equipped with more capable processing hardware and therefore exhibit lower per-packet service latency than ordinary relay nodes. The same physical path thus yields different uplink and downlink delays because of directional traffic characteristics and role-dependent processing, not because of asymmetric propagation physics.

2.3. Interdependent Coupling

The two layers are coupled through a one-to-one mapping between power buses and non-CC cyber nodes, encoded in a coupling matrix $A_{pc} \in \{0, 1\}^{|V_P| \times |V_C|}$.

Betweenness-assortative coupling. We construct A_{pc} by sorting power buses in descending order of their betweenness centrality in G_P, independently sorting non-CC cyber nodes in descending order of their betweenness centrality in G_C, and pairing them rank-by-rank. This assortative strategy ensures that structurally critical power buses are paired with structurally critical communication nodes, reflecting the engineering practice of provisioning important substations with high-quality communication resources [5].

Because each Monte Carlo trial generates a fresh BA communication network, a fresh CC selection, and a fresh coupling matrix, the reported metrics reflect averaged behavior across diverse cyber-physical configurations.

2.4. Capacity-Tolerance Model

Overload-driven cascading requires a model of component capacity. We adopt the standard tolerance-factor formulation:

Power branches:

$$C_e^P = (1 + \alpha) \cdot |P_e^{(0)}|, \tag{8}$$

where $P_e^{(0)}$ is the initial branch flow from DC power flow.

Cyber nodes:

$$C_i^C = (1 + \alpha) \cdot B_i^{(0)}, \tag{9}$$

where $B_i^{(0)}$ is the initial betweenness centrality of cyber node i.

Cyber edges:

$$C_e^{CE} = (1 + \alpha) \cdot B_e^{(0)}, \tag{10}$$

where $B_e^{(0)}$ is the initial edge betweenness of cyber link e.

The tolerance factor $\alpha \in [0, 1]$ governs excess capacity. At $\alpha = 0$ components operate at full capacity with no margin; at $\alpha = 1$ each component can absorb up to double its initial load. By sweeping α from 0 to 1 in increments of 0.1, we systematically explore how the interplay between structural margin and communication delay shapes cascade severity.

3. Delay-Integrated Cascading Failure Dynamics

The cascading failure model operates through a nested dual-loop architecture in which communication delay participates as an endogenous causal variable that actively shapes the cascade trajectory.

3.1. Attack Model

Each simulation begins with the targeted removal of a single non-CC cyber node—the non-CC node with the highest betweenness centrality in the initial communication network. This betweenness-targeted attack maximizes initial disruption to communication paths: the attacked node is a high-betweenness relay whose removal simultaneously lengthens many communication paths and may fragment the cyber network [1].

3.2. Structural Propagation—Inner Loop

Given the current set of failed nodes and edges, the inner loop iterates until no new structural failures emerge. Each iteration executes four steps:

Step 1: Cyber-layer island viability (FVC-ID). Connected components of the surviving cyber network are identified. Any component that does not contain at least one surviving control center is declared non-viable—without CC access, nodes can neither report measurements nor receive dispatch commands. All nodes and edges within non-viable components are marked as failed. This implements the Functional Viability-Constrained Island Detection (FVC-ID) mechanism introduced in our prior work [5].

Step 2: Forward propagation (cyber → power). Each newly failed non-CC cyber node triggers the failure of its coupled power bus with probability p.

Step 3: Power-layer island viability. Connected components of the surviving power network are identified. Any component that contains neither a generator nor a load bus is declared non-viable, and all its buses and branches are marked as failed.

Step 4: Reverse propagation (power → cyber). Each newly failed power bus triggers the failure of its coupled non-CC cyber node with probability p.

Convergence. Steps 1–4 repeat until the sets of failed cyber and power nodes remain unchanged. The bidirectional propagation with probability p is the mechanism through which damage crosses the layer boundary, creating cascading chains that alternate between layers until convergence [13, 15].

3.3. Overload-Driven Cascade Expansion—Outer Loop

After the inner loop stabilizes, the surviving network undergoes overload evaluation with delay injection.

Step 1: Cyber-layer overload check. Betweenness centrality is recomputed on the surviving cyber network. Any non-CC node whose updated betweenness exceeds its capacity $C_i^C = (1 + \alpha) \cdot B_i^{(0)}$ is removed. Similarly, any link whose edge betweenness exceeds $C_e^{CE} = (1 + \alpha) \cdot B_e^{(0)}$ is removed.

Step 2: Delay-aware generation adjustment. For each surviving generator, the framework computes a delay-dependent control efficiency factor η_i based on the current state of the communication network and scales the generator's output accordingly. This step—detailed in Section 3.4—is the architectural innovation that closes the feedback loop. The scaling occurs *before* the DC power flow, ensuring that delay's impact propagates through the electrical network.

Step 3: DC power flow and branch overload. A DC power flow is solved using the delay-adjusted generator outputs. Any branch whose post-redistribution flow exceeds $C_e^P = (1 + \alpha) \cdot |P_e^{(0)}|$ is tripped.

Step 4: Convergence test. If no new failures occur in Steps 1 and 3, the cascade terminates. Otherwise, the newly failed components are incorporated and the process returns to the inner loop (Section 3.2).

The feedback mechanism. In a conventional cascade model, every generator produces its full scheduled output regardless of communication quality. In the present framework, generators whose communication paths have lengthened produce less power (reduced η_i), and generators that have lost all CC access produce no power ($\eta_i = 0$). This output reduction alters branch flows: branches that would not overload under ideal control may now overload because reduced generation forces more aggressive redistribution onto surviving paths. These additional overloads cause further structural failures, which degrade the cyber topology, which increases delay. Each delay scenario therefore traces a unique cascade trajectory—a property fundamentally impossible in post-hoc delay models.

3.4. Delay-Aware Generator Dispatch

For each surviving generator i that remains online after structural propagation:

1. **Identify the coupled cyber node.** Generator i's power bus is mapped to its non-CC cyber node via A_{pc}.
2. **Find the nearest reachable CC.** On the surviving cyber network, the shortest delay-weighted path from the generator's cyber node to each surviving CC is computed. The CC with the minimum-delay path is selected. If no CC is reachable, the generator is classified as *unreachable*.
3. **Compute delays.** The total measurement delay and execution delay for generator i are

$$\tau_{m,i} = \tau_{\mathrm{PB}\to\mathrm{nonCC}} + T_{\mathrm{up}}(\pi_i), \tag{11}$$

$$\tau_{e,i} = \tau_{\mathrm{nonCC}\to\mathrm{PB}} + T_{\mathrm{down}}(\pi_i), \tag{12}$$

where $\tau_{\mathrm{PB}\to\mathrm{nonCC}}$ is the field-side measurement interface delay and $\tau_{\mathrm{nonCC}\to\mathrm{PB}}$ is the field-side command execution delay.
4. **Compute control efficiency.** The delay-to-control mapping converts the two delay components into a composite efficiency factor:

$$\eta_i = \min\big(1,\ \max\big(0,\ (1-k_m\tau_{m,i})(1-k_e\tau_{e,i})\big)\big), \tag{13}$$

where k_m and k_e are delay sensitivity coefficients.
5. **Scale generator output.** The actual output is

$$P_i^{\mathrm{actual}} = P_i^{\mathrm{ref}} \cdot \eta_i. \tag{14}$$

Unreachable generators. If no surviving CC is reachable, the generator receives no control commands: $P_i^{\mathrm{actual}} = 0$, $\eta_i = 0$. This is the most severe consequence of cyber-layer degradation—complete loss of controllability—and it occurs before the DC power flow, directly contributing to power imbalance and further overloads.

3.5. Delay Severity Scenarios

Five delay-severity scenarios are defined by coherently scaling a common baseline parameter set (Table 1). The scale factor is applied multiplicatively to all delay-producing parameters: packet sizes, link distances, all service delays, and field-side interface delays. The link data rate R, propagation speed V, and sensitivity coefficients k_m, k_e remain fixed.

The `no_delay` scenario (scale 0) zeros all delay terms; every reachable generator retains $\eta_i = 1$. Any performance difference between `no_delay` and other scenarios is therefore causally attributable to communication delay.

3.6. Resilience Metrics

R_1: Load retention ratio. Let L_{initial} denote the total pre-attack load and $L_{\mathrm{surviving}}$ the total load at surviving buses:

$$R_1 = \frac{L_{\mathrm{surviving}}}{L_{\mathrm{initial}}}. \tag{15}$$

Table 1
Delay severity scenarios.

Scenario	Scale factor	Interpretation
`no_delay`	0.0	Zero communication delay
`light`	0.5	Well-provisioned communication
`baseline`	1.0	Nominal delay parameterization
`medium`	1.5	Moderately degraded
`heavy`	2.0	Severely degraded

Because delay is embedded within the cascade process, delay's influence on R_1 manifests endogenously through divergent cascade trajectories.

R_3: Generator dispatch deviation. For the N generators surviving at the terminal state with $P_j^{\mathrm{ref}} > 0$:

$$R_3 = \sqrt{\frac{1}{N}\sum_{j=1}^{N}\left(\frac{P_j^{\mathrm{actual}} - P_j^{\mathrm{ref}}}{P_j^{\mathrm{ref}}}\right)^2}. \tag{16}$$

$R_3 = 0$ indicates perfect dispatch fidelity; higher values indicate greater deviation. Unreachable generators contribute the maximum relative deviation of 1.0.

These two metrics serve complementary diagnostic roles. R_1 answers "how much load survives?"—an aggregate measure. R_3 answers "how accurately are surviving generators executing their dispatch?"—a functional quality measure. A system may retain high R_1 yet suffer high R_3, a form of hidden degradation that load-retention metrics alone cannot detect.

4. Experimental Analysis

4.1. Experimental Setup

Test system. The physical layer is the IEEE 39-bus system (39 buses, 46 branches, 10 generators). The cyber layer is a BA network with $|V_C| = 47$ nodes (39 non-CC + 8 CC), grown from a seed of $m_0 = 4$ nodes with $m_e = 2$ attachment edges.

Monte Carlo design. For each of the 1 000 independent trials, a new BA network, a new CC set (via MSIS), and a new A_{pc} are generated. The power layer is shared across trials.

Parameter grid. Each trial is executed under all five delay scenarios at each of 11 tolerance levels ($\alpha \in \{0.0, 0.1, \ldots, 1.0\}$), producing $1{,}000 \times 5 \times 11 = 55{,}000$ independent cascade simulations.

The complete parameter set is given in Table 2.

Evaluation protocol. For each cascade simulation, round-by-round snapshots record all failed components, surviving topology, and the delay injection log (containing η_i, $\tau_{m,i}$, $\tau_{e,i}$, reachability status for every generator at every round). Final-state R_1 is computed from

Table 2
Simulation parameters.

Category	Parameter	Value
Power layer	Test case	IEEE 39-bus
	Buses / Branches / Gen.	39 / 46 / 10
Cyber layer	Total nodes $\lvert V_C \rvert$	47
	Control centers n_{cc}	8
	BA seed m_0 / Attach. m_e	4 / 2
Coupling	Strategy	Betweenness-assortative
	CC selection	MSIS ranking
Cascade	Propagation prob. p	0.3
	Tolerance α	0.0 : 0.1 : 1.0
	Attack mode	Betw.-targeted
Communi-cation	Uplink packet S_{up}	8 192 bits
	Downlink packet S_{down}	2 048 bits
	Link rate R	10 Mbps
	Prop. speed V	2×10^5 km/s
	Link distance d	1 km
Service delays	CC: tx / rx / fwd	3 / 4 / 3 ms
	Non-CC: tx / rx / fwd	12 / 9 / 6 ms
Delay-to-control	Meas. interface $\tau_{\mathsf{PB}\to\mathsf{nCC}}$	100 ms
	Exec. interface $\tau_{\mathsf{nCC}\to\mathsf{PB}}$	120 ms
	Meas. sensitivity k_m	0.80 s^{-1}
	Exec. sensitivity k_e	0.60 s^{-1}
Sampling	Trials per $(\alpha, \text{scen.})$	1 000

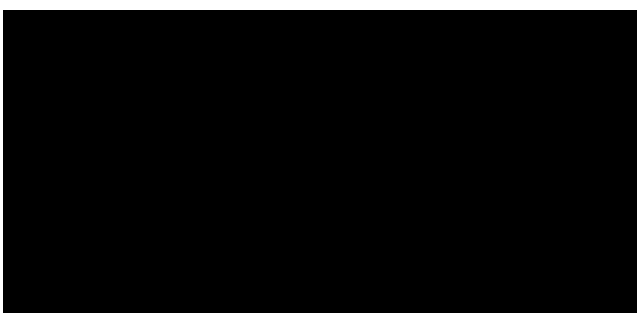

Figure 1: Mean load retention ratio R_1 versus tolerance factor α across five delay scenarios. The `no_delay` curve provides an upper bound; the inter-scenario gap widens at moderate-to-high α.

surviving bus loads; final-state R_3 from the terminal-round delay log. Trials that terminate early contribute their terminal R_1 to subsequent rounds via last-value-carried-forward (LVCF) padding.

4.2. Impact of Delay Severity on Load Retention

Fig. 1 presents the mean load retention ratio R_1 as a function of tolerance factor α across five delay scenarios, averaged over 1 000 trials.

Three patterns emerge. First, the `no_delay` curve lies consistently above all delay-present curves across the entire α range, confirming that zero-delay control provides an upper bound on system resilience.

Figure 2: Generator dispatch deviation R_3 versus α. All five curves are perfectly stratified (no crossings), with `heavy` at the top and `no_delay` at the bottom.

Second, the gap between `no_delay` and delay-present scenarios widens dramatically as α increases through the intermediate range. At $\alpha = 0.2$, all curves cluster near $R_1 \approx 0.20$—the system collapses regardless of delay. By $\alpha = 0.8$, the gap exceeds 0.25 in absolute terms. This widening reveals the *asymmetric vulnerability*: delay damage is most consequential when the power system has enough margin to survive without delay but not enough to absorb the additional overloads that delay creates.

Third, the transition from high to low R_1 steepens under heavier delay. Delay effectively narrows the operating window within which the system remains safe and shifts the critical transition point upward along the tolerance axis.

4.3. Generator Output Fidelity Under Delay

Fig. 2 presents R_3 across the same scenario–α grid.

The results are strikingly clean: all five curves are perfectly stratified with no crossings, maintaining strict monotonic ordering from `heavy` (worst fidelity) to `no_delay` (best fidelity).

R_3 *decreases* with increasing α—the opposite direction from R_1's increase. At higher α, fewer components overload, fewer generators lose reachability, and surviving generators retain higher η_i values. The inter-scenario separation persists even at high α where R_1 differences compress, revealing hidden degradation: a system that retains most load yet exhibits substantial control mismatch at the generator level. Such systems appear healthy by aggregate measures but are functionally stressed.

4.4. Distributional Analysis of Cascade Outcomes

Fig. 3 presents box plots of the R_1 distribution at three representative tolerance levels ($\alpha = 0.2, 0.5, 0.8$).

At low α ($= 0.2$), all scenarios produce broadly similar distributions—the system collapses regardless of delay. At moderate α ($= 0.5$), a striking divergence

Figure 3: Box plots of R_1 distribution at $\alpha = 0.2$, 0.5, and 0.8. Delay amplifies distributional spread and introduces a catastrophic-collapse tail absent from the `no_delay` case.

Figure 4: Delay penalty ΔR_1 heatmap. The penalty ignites at Round 2, amplifies through Rounds 3–5, and saturates thereafter. Low α remains dark (floor effect); the largest penalty concentrates at moderate-to-high α.

emerges: the `no_delay` distribution shifts upward while delay-present scenarios develop lower whiskers extending toward $R_1 \approx 0.05$, revealing a tail of near-total collapse absent from `no_delay`. At high α ($= 0.8$), `no_delay` concentrates near $R_1 \approx 1$ while delay-present scenarios remain dispersed in the 0.40–0.60 range.

The mechanism underlying this variability amplification is the interaction between random cyber topology and the delay feedback loop. Under ideal communication, topological differences affect only which nodes fail structurally. Under heavy delay, topological differences translate into different η_i profiles, different overload patterns, and divergent cascade trajectories. Small differences in cyber connectivity, inconsequential under ideal communication, become decisive under heavy delay.

4.5. Temporal-Spatial Structure of the Delay Penalty

Define the round-level delay penalty:

$$\Delta R_1(r, \alpha) = R_1^{\text{no}-\text{delay}}(r, \alpha) - R_1^{\text{heavy}}(r, \alpha). \quad (17)$$

Fig. 4 presents ΔR_1 as a heatmap over the (cascade round, α) plane.

The heatmap reveals a striking temporal-spatial structure:

Round 1 is delay-neutral. The leftmost column is uniformly dark ($\Delta R_1 \approx 0$): Round 1 reflects the immediate structural consequences of the initial attack, identical across delay scenarios.

The delay penalty ignites at Round 2. A sharp transition occurs at $\alpha \geq 0.4$: ΔR_1 leaps from near zero to substantial values in a single round. This is the moment the feedback loop first fires.

Amplification through Rounds 3–5. The penalty intensifies as each round of delay-induced overloads creates further structural damage, producing compounding rather than linear accumulation.

Saturation after Round 5. Colors stabilize: 80–90% of avoidable delay damage is locked in by the fifth round.

Low α remains dark. When physical margins are so tight that even the delay-free system collapses, there is no room for delay to widen the gap.

Operational interpretation. An operator who could mitigate communication delay would achieve maximum benefit by acting at Round 2—before the feedback loop amplifies the initial penalty into compounding damage. By Round 5, the intervention window has largely closed.

Table 3
Safety-level classification.

Level	R_1 range	Interpretation
Green	$> 85\%$	Normal operation
Yellow	70%–85%	Caution required
Orange	50%–70%	High risk
Red	$< 50\%$	System collapse

4.6. Safety-Level Classification

Based on R_1, each (α, scenario) outcome is mapped to an operational safety level (Table 3).

The classification reveals that delay shifts the safety boundaries. A tolerance level yielding Green under `no_delay` may yield Yellow or Orange under `heavy` delay—not because the system is structurally different, but because delay triggers additional cascade rounds that the delay-free analysis would not predict. Safety margins computed without accounting for communication delay are therefore potentially non-conservative.

4.7. Sensitivity Analysis of Delay-Mitigation Actions

Five candidate engineering actions are evaluated, each modifying a single delay parameter under the `heavy` scenario while all others remain at their heavy-delay values (Table 4).

For each action, the full Monte Carlo is repeated across all α values. The resulting R_1 and R_3 distributions are compared against both the unmodified `heavy` scenario and the `no_delay` benchmark to quantify each action's marginal benefit in absolute terms (ΔR_1) and as a fraction of the maximum recoverable penalty.

Table 4
Delay-mitigation actions.

Action	Description	Parameter change
A1	Bandwidth upgrade	R: 10 → 100 Mbps
A2	Endpoint optimization	CC/non-CC tx/rx halved
A3	Forwarding optimization	CC/non-CC fwd halved
A4	Meas. interface speedup	100 → 20 ms
A5	Exec. interface speedup	120 → 30 ms

Note: Detailed numerical results will be incorporated once the simulation completes.

4.8. Timing-Action Interaction Experiment

The heatmap analysis identified Round 2 as the critical intervention window and $\alpha = 0.5$–0.6 as the most delay-sensitive operating regime. A 2×2 factorial experiment crosses timing (correct: Round 2; incorrect: Round 5) with action quality (best vs. worst action from Section 4.7). This tests whether the timing-action interaction is additive or synergistic.

Note: Detailed numerical results will be incorporated once the simulation completes.

5. Conclusion

This paper constructs a delay-integrated cascading failure framework for cyber-physical power systems in which communication delay is not an after-the-fact observation but an endogenous causal variable embedded within the cascade process. At each cascade round, every surviving generator's output is scaled by a delay-dependent control efficiency factor η_i before the DC power flow that determines overloads. This closes a feedback loop that prior models leave open: delay reduces generation, altered power flows cause additional overloads, structural failures degrade the communication topology, and the degraded topology amplifies delay for subsequent rounds.

The framework models communication delay with full physical grounding—decomposing end-to-end latency into serialization, propagation, endpoint service, and forwarding components with directional asymmetry between uplink measurement traffic and downlink command traffic. Control centers are placed via a multi-attribute structural importance score (MSIS), and interdependent coupling uses a betweenness-assortative strategy that pairs structurally critical power buses with structurally critical communication nodes.

Monte Carlo simulation across 1 000 independent coupling realizations, 11 tolerance levels, and 5 delay scenarios on the IEEE 39-bus system yields four principal findings:

1. **Genuinely different cascade trajectories.** Each delay scenario traces a unique path through the failure space—different overload patterns, failure sequences, and terminal states. This qualitative divergence is fundamentally impossible in post-hoc delay models.
2. **Asymmetrically concentrated delay penalty.** The ΔR_1 heatmap reveals that delay damage is largest at moderate-to-high tolerance margins and late cascade rounds—precisely where delay-free analysis predicts safety. Delay-free models are most misleading where operators are most likely to rely on them.
3. **Amplified stochastic variability.** Under heavy delay, the R_1 distribution widens substantially, introducing a catastrophic-collapse tail absent from the ideal-communication case.
4. **Hidden functional degradation.** R_3 maintains clean stratification across delay scenarios even at high α where R_1 differences compress, exposing a functional quality deficit invisible to aggregate metrics.

The practical implication is direct: robustness assessment and safety-margin setting for CPPS must incorporate communication delay as a causal variable within the cascade model. Defense investments should prioritize maintaining communication reachability—through path redundancy, CC diversity, and connectivity-preserving reinforcement—over minimizing per-hop latency, because complete loss of controllability ($\eta = 0$) produces qualitatively more severe damage than degraded controllability ($0 < \eta < 1$). The round-level analysis further suggests that delay-mitigation interventions are most effective early in the cascade (Round 2) before the feedback loop compounds initial damage into irreversible collapse.

Future work will extend this framework in three directions: (i) a nonlinear or threshold-based delay-to-control mapping that captures stability loss beyond critical delay values; (ii) generalization to larger test systems (IEEE 118-bus) and alternative communication topologies; and (iii) delay-aware defense design that explicitly targets the reachability-dominated fragmentation threshold.

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